Minimalist Foundations

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Course outline

Monday	From X-bar theory to Bare Phrase Structure: The emergence of Merge
Tuesday	Merge, Move, and Agree
Wednesday	The expansion of Merge: External / Internal Merge and Phase theory
Thursday	The Strong Minimalist Thesis, Merge, and the problem of labelling
Friday	The Miracle Creed: Can syntax be reduced to Merge?

The Strong Minimalist Thesis, Merge, and the problem of labelling

Outline

1. The Strong Minimalist Thesis
2. Towards a Labelling Algorithm
3. Internal Merge and symmetry breaking
4. Summary and outlook

The Strong Minimalist Thesis, Merge, and the problem of labelling

The Strong Minimalist Thesis

- Chomsky (2000) gives a name to the foundational agenda of Minimalism:
  The Strong Minimalist Thesis (SMT) treats the faculty of language as an optimally designed computational system that contains no more machinery than is needed to generate representations that can be 'read' by the conceptual-intentional and sensory-motor systems.

(1) 'Language is an optimal solution to legibility conditions.'
(Chomsky 2000: 96, ex. (2))

The Strong Minimalist Thesis, Merge, and the problem of labelling

The Strong Minimalist Thesis

‘Adopting [the SMT] as a point of departure, assume that FL provides no machinery beyond what is needed to satisfy minimal requirements of legibility and that it functions in as simple a way as possible. We would like to establish such conclusions as these:

a. The only linguistically significant levels are the interface levels.

b. The *Interpretability Condition*: LIs have no features other than those interpreted at the interface, properties of sound and meaning.

c. The *Inclusiveness Condition*: No new features are introduced by C_{HL}.

d. Relations that enter into C_{HL} either (i) are imposed by legibility conditions or (ii) fall out in some natural way from the computational process.’

(Chomsky 2000: 112-113).

The Strong Minimalist Thesis, Merge, and the problem of labelling

Towards a Labelling Algorithm

- Early Minimalist approaches treated labelling as an inherent part of Merge.

> ‘Applied to two objects α and β, Merge forms the new object K, eliminating α and β … The simplest object constructed from α and β is the set {α , β}, so we take K to involve at least this set, where α and β are the constituents of K. Does that suffice? Output conditions dictate otherwise; thus, verbal and nominal elements are interpreted differently at LF and behave differently in the phonological component. K must therefore at least (and we assume at most) be of the form {γ, {α , β}}, where γ identifies the type to which K belongs, indicating its relevant properties. Call γ the label of K’ (Chomsky 1995: 223).

(2) [rebel [against the government]]

(3) a. V, a rebel against the government who always wears a Guy Fawkes mask …

b. The students will rebel against the government.

The Strong Minimalist Thesis, Merge, and the problem of labelling

Towards a Labelling Algorithm

- Given the SMT, the assumption that Merge intrinsically assigns labels becomes a natural target for reconsideration. Rather than stipulating labels as part of syntactic computation, the question becomes whether they can be derived from more general properties of the computational system.

The Strong Minimalist Thesis, Merge, and the problem of labelling

Towards a Labelling Algorithm

- Chomsky (2005) suggests that **specifying labels as part of the syntactic objects created by Merge may be redundant**, raising the question of whether labels can instead be derived from independently necessary properties of the computational system.

  'Each syntactic object generated contains information relevant to further computation. Optimally, that will be captured entirely in a single designated element, which should furthermore be identifiable with minimal search: its label, the element taken to be "projected'' in X-bar-theoretic systems. The label, which will invariably be a lexical item introduced by external Merge, should be the sole probe for operations internal to the syntactic object, and the only element visible for further computations. We would hope that labels are determined by a natural algorithm ...

  Note that labels, or some counterpart, are the minimum of what is required, on the very weak assumption that at least some information about a syntactic object is needed for further computation, both for search within it and for its external role. Labeling underlies a variety of asymmetries: for example, in a head-XP construction, the label will always be the head, and the XP a "dependency'' ... Therefore, such asymmetries need not be specified in the syntactic object itself, and must not be, because they are **redundant**'
  (Chomsky 2005: 14).

The Strong Minimalist Thesis, Merge, and the problem of labelling

Towards a Labelling Algorithm

- Chomsky (2013) proposes that labels should be determined by a separate **labelling algorithm** operating over the structure created by Merge.

> ‘Projection is a theory-internal notion, part of the computational process G[enerative]P[rocedure]. For a syntactic object SO to be interpreted, some information is necessary about it: what kind of object is it? Labeling is the process of providing that information. Under P[hrase]S[tructure]G[rammar] and its offshoots, labeling is part of the process of forming a syntactic object SO. But that is no longer true when the stipulations of these systems are eliminated in the simpler Merge-based conception of UG. We assume, then, that there is a fixed labeling algorithm LA that licenses SOs so that they can be interpreted at the interfaces, operating at the phase level along with other operations.[30]’

The Strong Minimalist Thesis, Merge, and the problem of labelling

Towards a Labelling Algorithm

- This labelling algorithm is suggested to derive from a Minimal Search procedure, which Chomsky presents as an instance of the kind of general principle of efficient computation that, under the third-factor perspective, may reduce the need for language-specific stipulations.

> ‘The simplest assumption is that LA is just **minimal search**, presumably appropriating a **third factor principle**, as in Agree and other operations. In the best case, the relevant information about SO will be provided by a single designated element within it: a computational atom, to first approximation a lexical item LI, a head. This LI should provide the label found by LA, when the algorithm can apply’
> (Chomsky 2013: 43; emphasis added).

The Strong Minimalist Thesis, Merge, and the problem of labelling

Towards a Labelling Algorithm

- The ‘third-factor perspective’ is introduced in Chomsky (2005), which re-conceptualises the factors involved in language acquisition and the design of the language faculty.

‘Assuming that the faculty of language has the general properties of other biological systems, we should, therefore, be seeking **three factors** that enter into the growth of language in the individual:

- Genetic endowment ...
- Experience ...
- **Principles not specific to the faculty of language**.

The third factor falls into several subtypes: (a) principles of data analysis that might be used in language acquisition and other domains; (b) principles of structural architecture and developmental constraints that enter into canalization, organic form, and action over a wide range, including **principles of efficient computation, which would be expected to be of particular significance for computational systems such as language**. It is the second of these subcategories that should be of particular significance in determining the nature of attainable languages’
(Chomsky 2005: 6; emphasis added).

The Strong Minimalist Thesis, Merge, and the problem of labelling

Towards a Labelling Algorithm

- Minimal Search is the principle that, when the syntactic computation searches a domain for a target, it identifies the closest structurally accessible element that satisfies the relevant condition and searches no further. The underlying idea is that syntactic computation should minimise search by avoiding unnecessary inspection of more deeply embedded structure.

- Never formally defined (by Chomsky, but cf. Ke 2024), the notion first appears in Chomsky (2000: 99):

  '… we might inquire into the complexity of the generative procedure … One category concerns "least effort" conditions … Another category seeks to **reduce "search space" for computation**: "Shortest Movement/Attract", successive-cyclic movement (Relativized Minimality, Subjacency), restriction of search to c-command or minimal domains, and so on' .

- Chomsky (2004: 109) first mentions Minimal Search in the context of (Internal) Merge:

  'Elementary considerations of efficient computation require that Merge of α to β involves **minimal search** of β to determine where α is introduced, as well as least tampering with β: search therefore satisfies some locality condition (let us say, defined by least embedding, "closest" under c-command), and Merge satisfies an extension condition'.

The Strong Minimalist Thesis, Merge, and the problem of labelling

Internal Merge and symmetry breaking

- The proposal that Merge should not assign labels is motivated not only by conceptual considerations arising from the Strong Minimalist Thesis, but also by empirical phenomena involving configurations in which a unique label cannot straightforwardly be determined by projection from a head.

Chomsky (2013: 43):

We assume, then, that there is a fixed labeling algorithm LA that licenses SOs so that they can be interpreted at the interfaces, operating at the phase level along with other operations.[30]

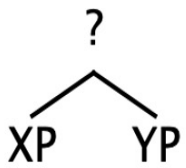
[30] Note that this is a modification of earlier proposals (of mine in particular) that labeling is a prerequisite for entering into computation. **That cannot be … or it would block many cases of EM, e.g., Merge (Z, {XP, YP}) = {Z, {XP, YP}}**'.

The Strong Minimalist Thesis, Merge, and the problem of labelling

Internal Merge and symmetry breaking

- The suggestion made in the footnote is that it is possible for External Merge to apply to a syntactic object that does not have a label.

(4) a.



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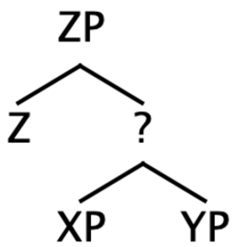


The Strong Minimalist Thesis, Merge, and the problem of labelling

Internal Merge and symmetry breaking

- Do we have any empirical evidence that labelling can be delayed?
- Moro (1997) argues for **copular constructions** to be derived by either the subject or the predicative noun phrase of a small clause raising to the sentence's SpecIP position.

(5) a. A picture of the wall was the cause of the riot.

b. The cause of the riot was a picture of the wall.

(Moro 1997: 35, ex. (42))

(6) [was [a picture of the wall the cause of the riot]]

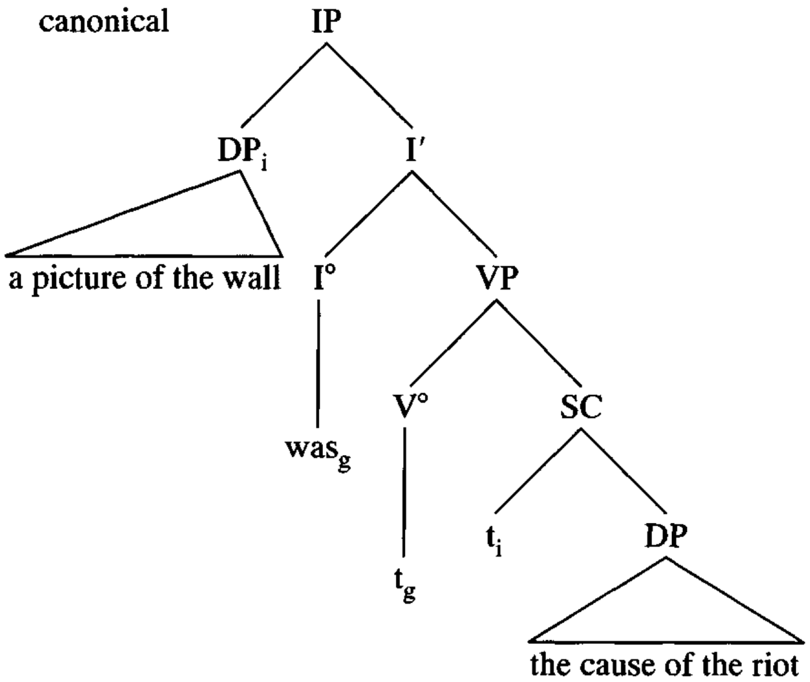
(7) a. I consider [a picture of the wall the cause of the riot].

b. [The cause of the riot a picture of the wall]? I can't believe it!
(cf. Moro 1997: 31, ex. (38a) & (38c))

The Strong Minimalist Thesis, Merge, and the problem of labelling

Internal Merge and symmetry breaking

(8) a. A picture of the wall was the cause of the riot.

b. 

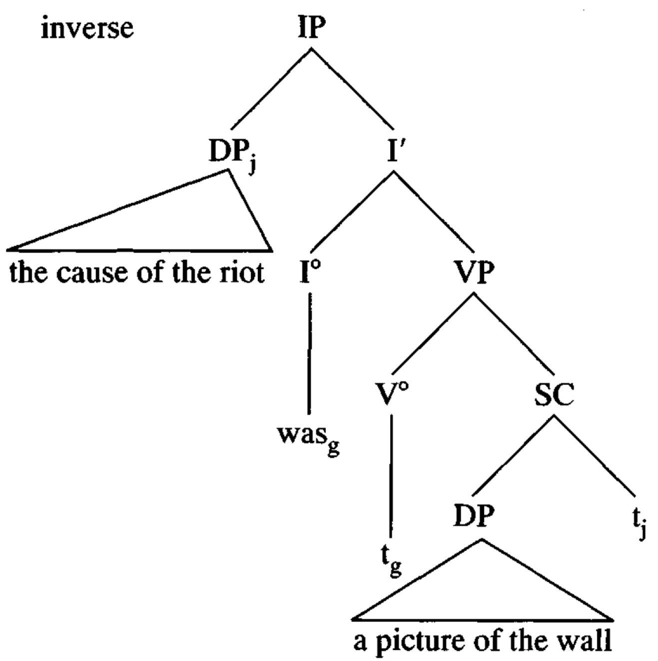
(Moro 1997: 35, ex. (43a))

The Strong Minimalist Thesis, Merge, and the problem of labelling

Internal Merge and symmetry breaking

(9) a. The cause of the riot was a picture on the wall.

b.



(Moro 1997: 35, ex. (43b))

The Strong Minimalist Thesis, Merge, and the problem of labelling

Internal Merge and symmetry breaking

- Moro (2000) proposes that symmetric configurations may be generated during the derivation but must be rendered asymmetric before linearisation. Movement functions as the mechanism that breaks the symmetry.

- Chomsky (2013) reinterprets Moro’s notion of ‘dynamic antisymmetry’ (in which movement breaks symmetry to satisfy the requirements of linearisation) in terms of the Labelling Algorithm.

The Strong Minimalist Thesis, Merge, and the problem of labelling

Internal Merge and symmetry breaking

- **Labelling Algorithm** (Chomsky 2013: 43):

  Suppose SO = {H, XP}, H a head and XP not a head. Then LA will select H as the label, and the usual procedures of interpretation at the interfaces can proceed.

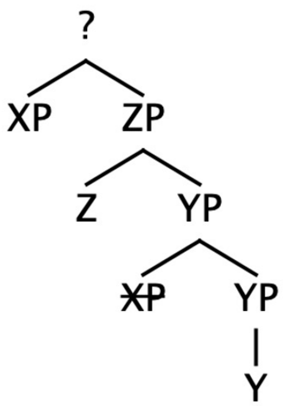
  The interesting case is SO = {XP, YP}, neither a head ... Here minimal search is ambiguous, locating the heads X, Y of XP, YP, respectively. There are, then, two ways in which SO can be labeled: (A) modify SO so that there is only one visible head, or (B) X and Y are identical in a relevant respect, providing the same label, which can be taken as the label of the SO.

The Strong Minimalist Thesis, Merge, and the problem of labelling

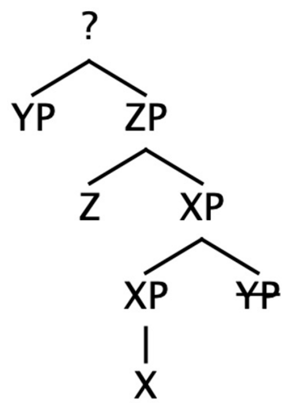
Internal Merge and symmetry breaking

- The Internal Merge of either the small-clause subject (XP) or predicate (YP) at the root of the tree instantiates **option (A) for resolving the labelling problem** posed by SO = {XP, YP}. If either XP or YP is raised, the small-clause structure is modified so that only one of the two elements remains visible to Minimal Search, thereby allowing the small-clause SO to be labelled.

(10) a.



b.

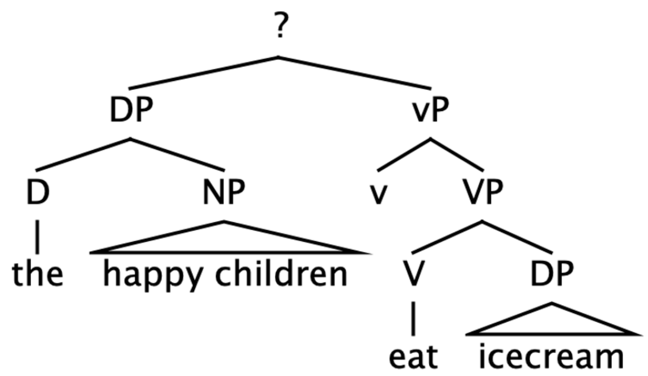


The Strong Minimalist Thesis, Merge, and the problem of labelling

Internal Merge and symmetry breaking

- '... or **(B)** X and Y are identical in a relevant respect, providing the same label, which can be taken as the label of the SO' (Chomsky 2013: 43)'

(11)



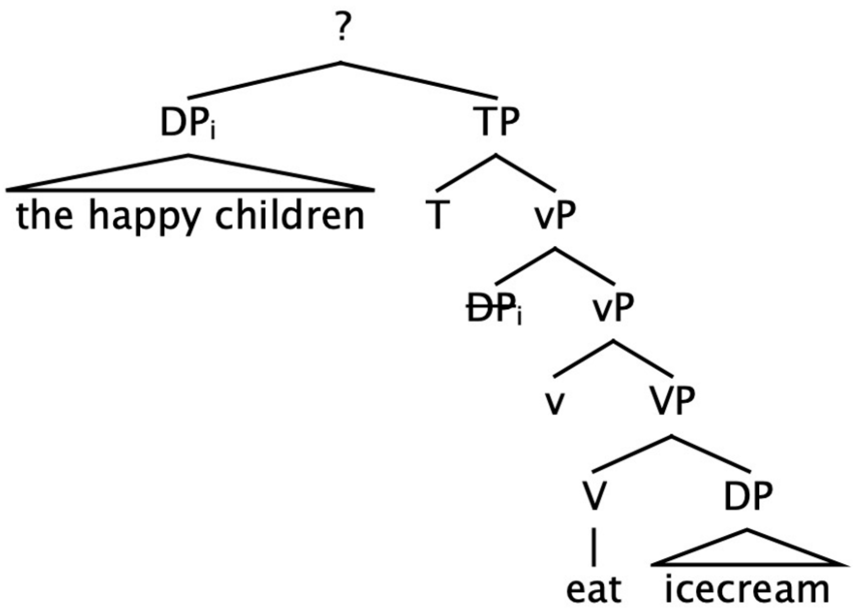
- Neither DP nor *v*P can determine the label of the SO created by their Merge, since Minimal Search finds the relevant heads of both phrases at the same depth.

The Strong Minimalist Thesis, Merge, and the problem of labelling

Internal Merge and symmetry breaking

- The need for labelling forces Internal Merge of the subject DP into SpecTP, but this does not eliminate the labelling problem.

(12)

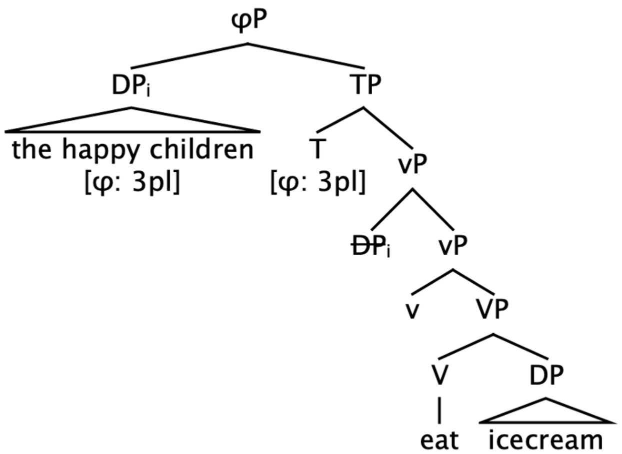


The Strong Minimalist Thesis, Merge, and the problem of labelling

Internal Merge and symmetry breaking

- Chomsky (2013: 45) proposes that the resulting configuration can instead be labelled through the sharing of ϕ-features between the subject DP and TP.

(13)



- Labels therefore need not always be categories projected from heads. In an otherwise problematic {XP, YP} configuration, the label can instead be a property shared by the two phrases.

The Strong Minimalist Thesis, Merge, and the problem of labelling

Internal Merge and symmetry breaking

- *Try your hand at some syntactic analysis*

- The phase-theoretic analysis of a *wh*-question like (14) postulates that the *wh*-phrase must move successive-cyclically through the phase edges, first to SpecvP and then to SpecCP.

(14) Who did Mary see?

Consider the individual steps in the derivation of (14). Do the resulting syntactic objects pose labelling problems, and if so, how does the computational system resolve them?

The Strong Minimalist Thesis, Merge, and the problem of labelling

Summary and Outlook

- The Strong Minimalist Thesis, which understands the language faculty as an optimal solution to the legibility conditions imposed by the interface levels, motivates removing labelling from the operation Merge, since Merge should contain no machinery beyond what is required to construct syntactic objects that can satisfy those interface conditions.
- Once labelling is derived by an independent Labelling Algorithm, the requirement that syntactic objects be labelled can itself motivate Internal Merge. Where Merge creates a structure that the Labelling Algorithm cannot label, Internal Merge can break the symmetry of the configuration, thereby modifying the structure in a way that makes a label identifiable.

- Chomsky's (2013) Labelling Algorithm raises a broader question about the role of labelling in the syntax: can requirements on the identification of labels provide a general motivation for Internal Merge?

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